

Algebra 1 – Unit 4 Test A

Name: _____

Evaluate each function. Show your work.

1. $y = 2x + 3$ when $x = 4$

2. $f(x) = -3x + 1$ when $x = -2$

Find the slope. Show your work.

3. $(2, 5)$ and $(6, 13)$

Identify the slope type (positive, negative, zero, or undefined). No work needed.

4. $(4, 1)$ and $(4, -5)$ _____

5. $(-3, 2)$ and $(5, 2)$ _____

Answer each question. Show your work.

6. Find the x -intercept and y -intercept of $y = 2x - 4$.

Write an equation for each line. Show your work.

7. Slope $m = 3$, through $(0, -2)$

8. Through $(0, 4)$ and $(2, 8)$

Solve each system. Show your work.

9.
$$\begin{cases} y = 2x - 1 \\ y = -x + 5 \end{cases}$$

10.
$$\begin{cases} x + y = 8 \\ x - y = 2 \end{cases}$$

Real-World Problems

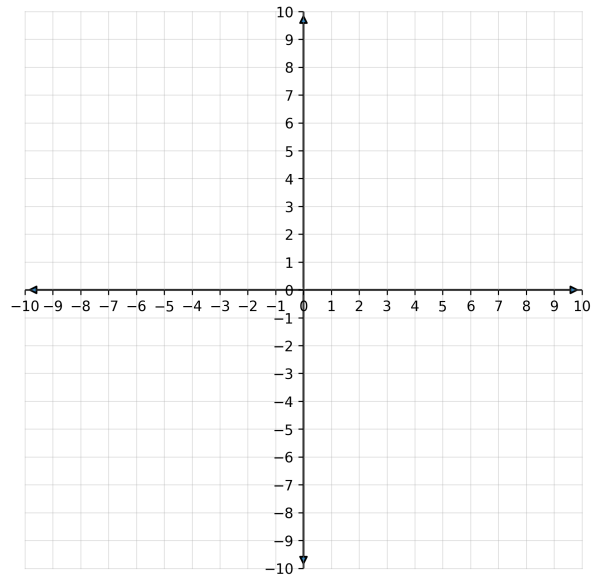
11. A car is worth \$12,000 when purchased and loses \$1,500 per year. Write an equation for its value y after x years. What do the slope and y -intercept each mean in this situation?

12. Using your equation from problem 11, what is the car worth after 4 years?

Complete the table of values and graph the line.

13. $y = -\frac{1}{2}x + 2$

x	y
-2	
0	
2	
4	



Graph the solution region.

14. You are selling candles for \$5 each and soaps for \$3 each. You need to earn at least \$30.
Write an inequality for this situation.

15. Graph the solution region from problem 14. Label your axes.

